

Reciprocal event audit

Signal Space / GROSS Test 2

Scientific classification: pass. Technical execution: completed.

Question: does the reciprocal event law preserve its declared ledger and causal scheduling?

Method: four seeded preparations, $\lambda=0$ and 0.1 , $\kappa=\pi/2$; twelve acyclic gates per circuit.

DOP853 numerical evolution is compared with half integration tolerances and a separate closed matrix-exponential solution.

Variables: z_1, z_2 are complex wave spinors; w is unit memory; $P=ww^\dagger$, $d=z_1-z_2$, $N=|z_1|^2+|z_2|^2$.

The gate Hamiltonian is $H=\kappa |w^\dagger d|^2 + \lambda N^2/2$. All quantities and the gate parameter s are dimensionless.

The exact reference uses the constant Hermitian matrix $S=dd^\dagger+2ww^\dagger$. The protocol derives this solution and defines all normalization and error gates.

Largest conservation residual: $3.50812e-10$. Exact-reference difference: $2.45401e-11$. Outside-ancestor Jacobian: 0.

Run: run-81319816afe7cf8e. Analysis: analysis-0001-8c6b8920.

Solver revision: 183b95feeae1cc1d4bd79ad3b3cd506f4952fba0. Config: e0caf90349d6a04a2f068137e4fabccccbabfae5429fcb8ff5ef16ff1e2e3f00.

Error controls: within-gate norm, matrix-charge and Hamiltonian ledgers; complete-cut charge; half-tolerance integration; closed solution; central-difference step halving; local frames and valid topological rescheduling.

Boundary sensitivity: no spatial box or mesh exists in this test. All circuit wires are retained, and the disconnected branch supplies a causal null. Changing incidence would change the apparatus, not refine it.

Limits: causal support is not a metric. No clock, full mode spectrum, direction-dependent propagation or emergent spacetime has been evaluated. Event H is not asserted to be a globally conserved circuit energy.

Next if accepted: Test 3 exact-router dispersion, with orthogonal, oblique and collinear triads, reversed order and full-zone inspection. Retain physical memory modes for Test 4 and directional comparisons for Test 11.

causal-incidence

Question: Do event outputs depend only on their causal ancestors?

Reading: Green events descend from initial wave 0. Gray heatmap cells are nonancestors; their largest response is 0. Step-halving Jacobian difference is $4.24e-08$.

Significance: Tests participant-only causal dependence and includes a disconnected branch as a null.

Nonzero allowed responses keep the test from passing through absent dynamics.

Limitation: Incidence is supplied, not derived space. Jacobians cover one preparation, both couplings and physical memory tangents; no universal metric is inferred.

matrix-balance

Question: Does reciprocal memory exchange close the matrix-charge ledger?

Reading: Wave and memory charge components exchange with opposite signs. The smallest frozen-memory relative charge defect across all cases is 0.198.

Significance: Tests whether reciprocal memory dynamics closes the conserved matrix ledger; a prescribed frozen projector is an externally driven control.

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Limitation: Displayed trajectory is the first gate of preparation 0. The Pauli component is basis dependent; the full Frobenius residual is the acceptance observable. This is not mechanical recoil.

numerical-audit

Question: Do conservation and equivalent descriptions survive numerical refinement?

Reading: Largest event conservation residual $3.51\text{e-}10$; exact-solution difference $2.45\text{e-}11$; tolerance-halving difference $2.14\text{e-}11$. Compare with $1\text{e-}9$.

Significance: Checks integrated dynamics against a closed solution and tests description invariance under legal schedules and local frames.

Limitation: Finite circuit and finite states. $U(2)$ covariance is not Lorentz covariance; numerical tolerance refinement is not a continuum limit. Exact zeros use a display floor.

order-controls

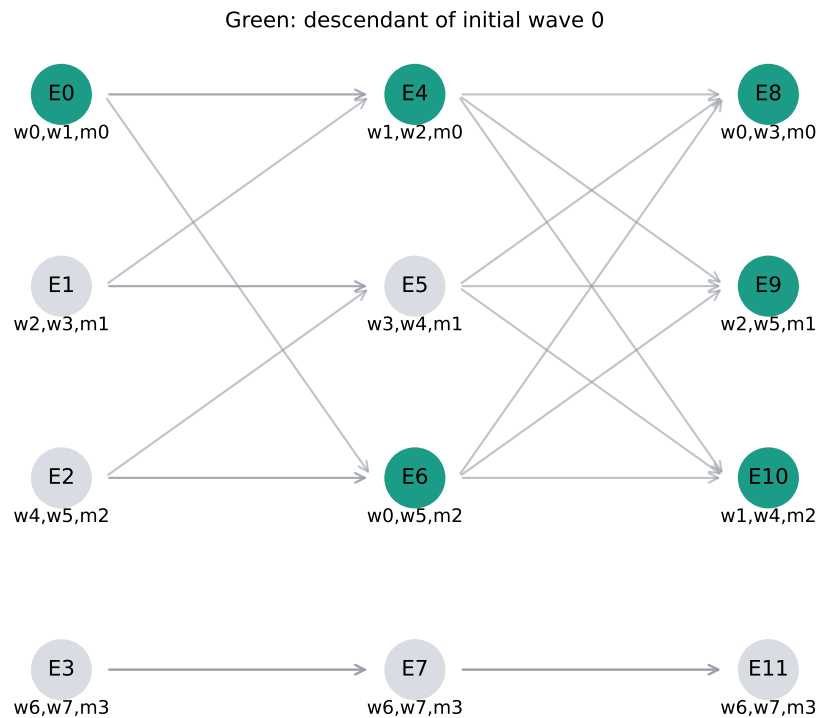
Question: Can the audit distinguish physical interventions from rescheduling?

Reading: Smallest shared-memory projector change 0.0673; smallest frozen-memory defect 0.198. Every preparation is compared with $1\text{e-}4$.

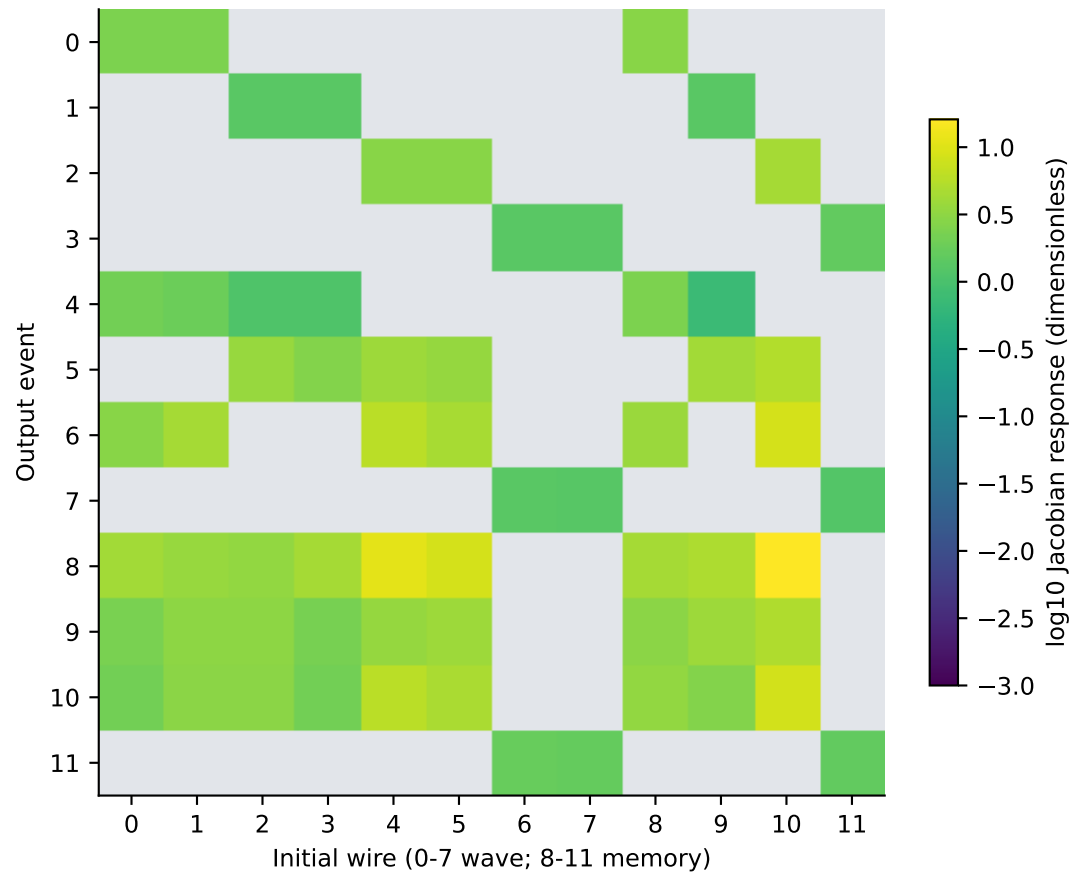
Significance: Distinguishes changes to physical event ordering and reciprocal feedback from permissible rescheduling of independent events.

Limitation: Generic noncommutativity is expected, not guaranteed for all states. These fixed controls do not define new candidate models or test propagation.

Prescribed incidence (arrows are wires)



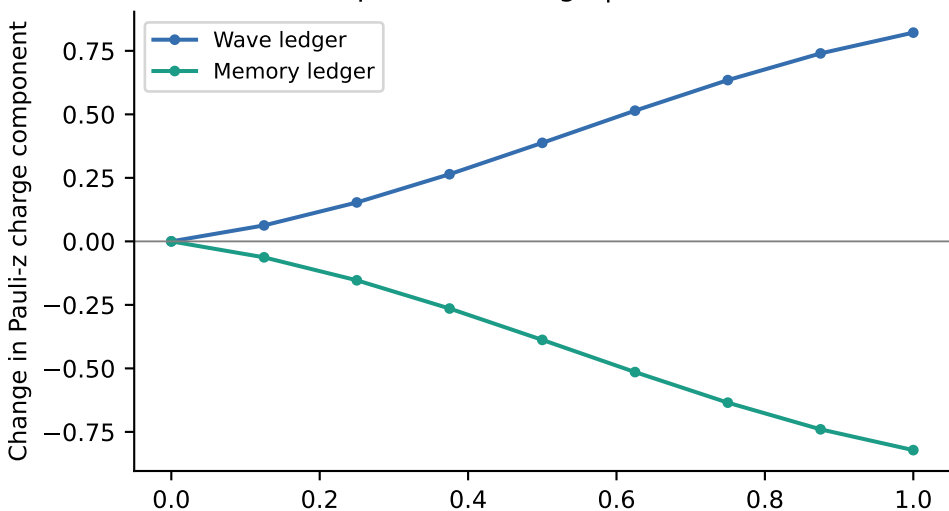
Measured causal response



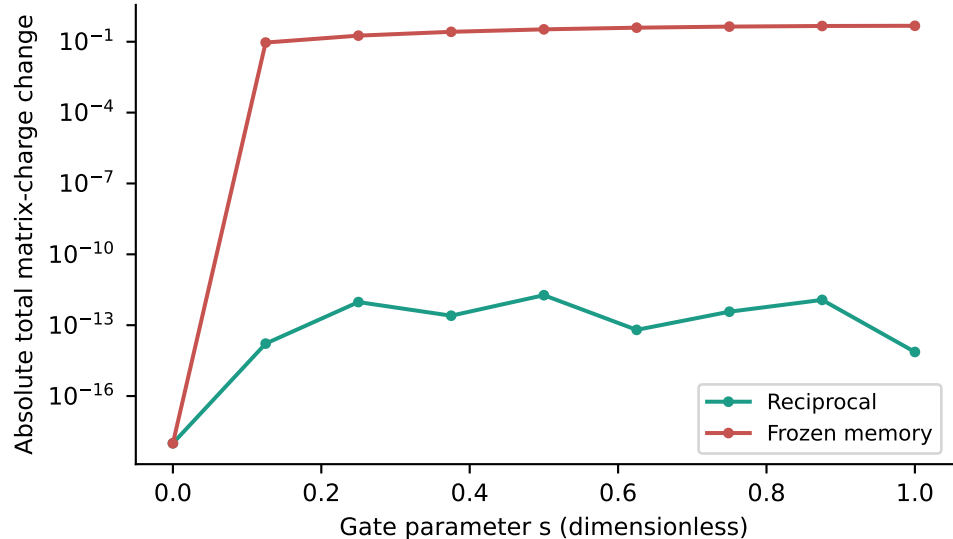
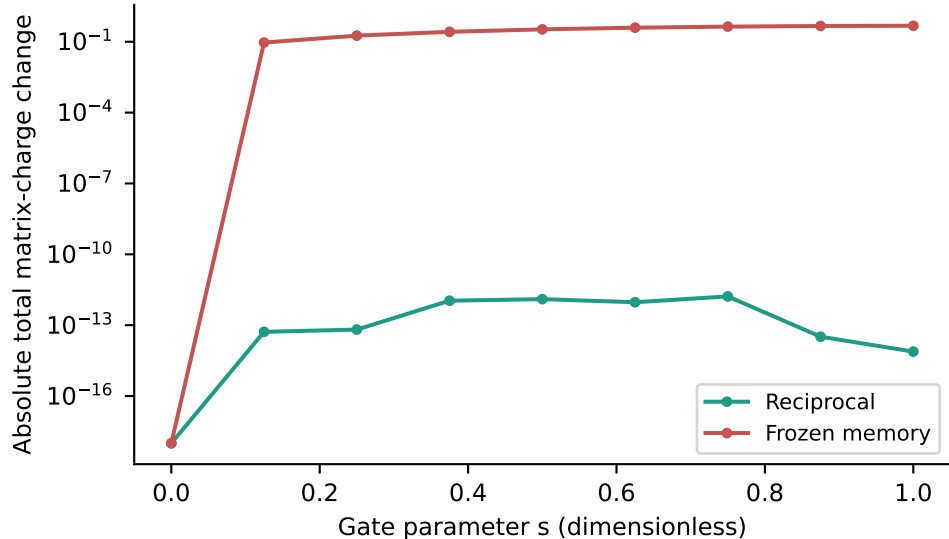
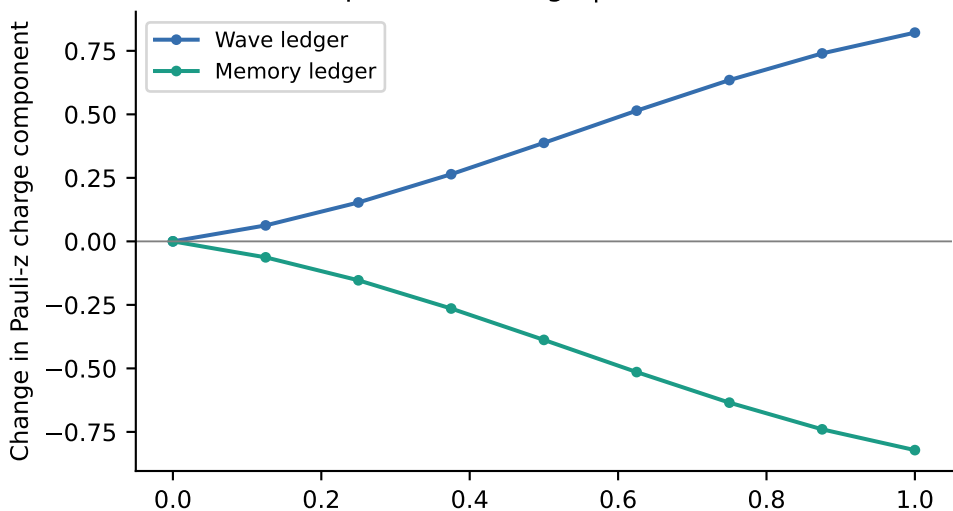
Gray: nonancestor; maximum response 0

Test 2 | Backreaction closes the ledger; freezing memory opens it

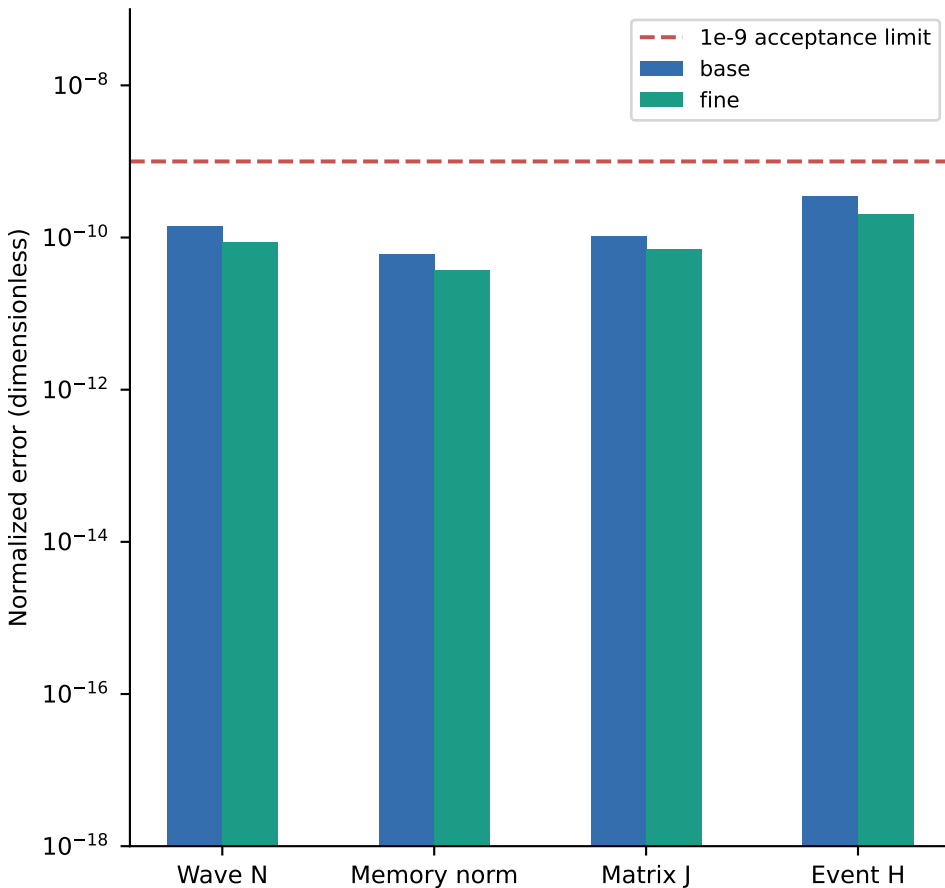
Reciprocal exchange | $\lambda = 0$



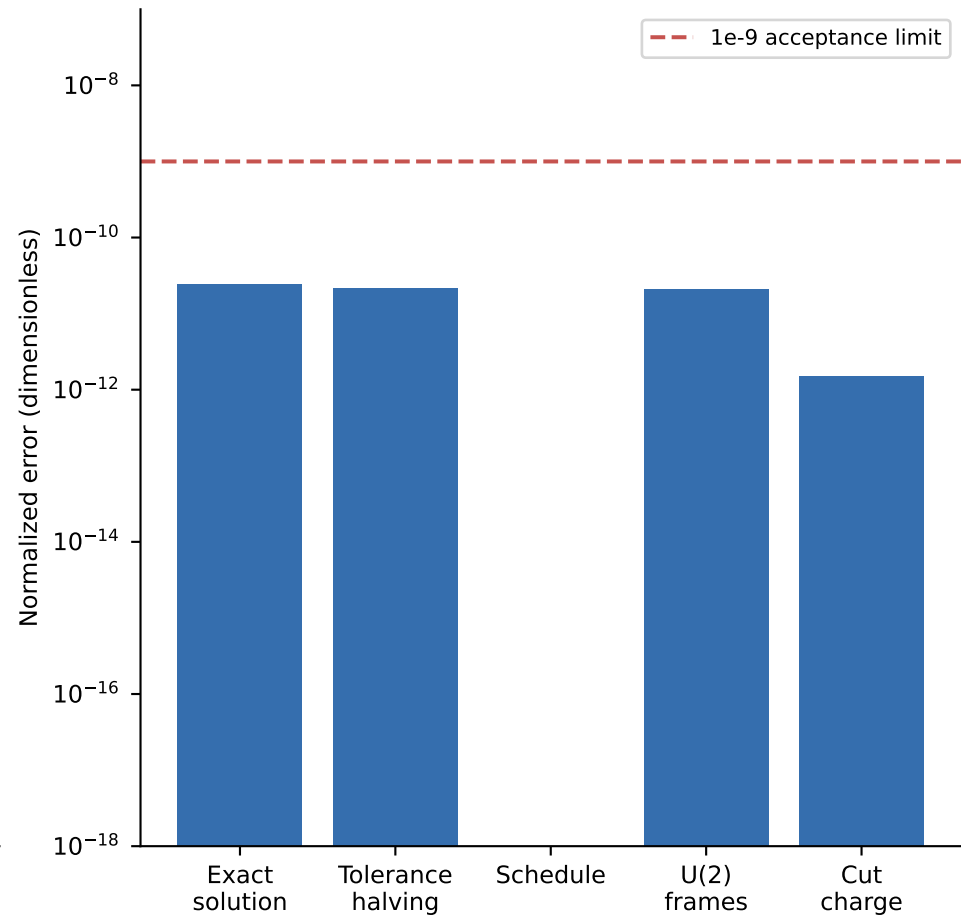
Reciprocal exchange | $\lambda = 0.1$



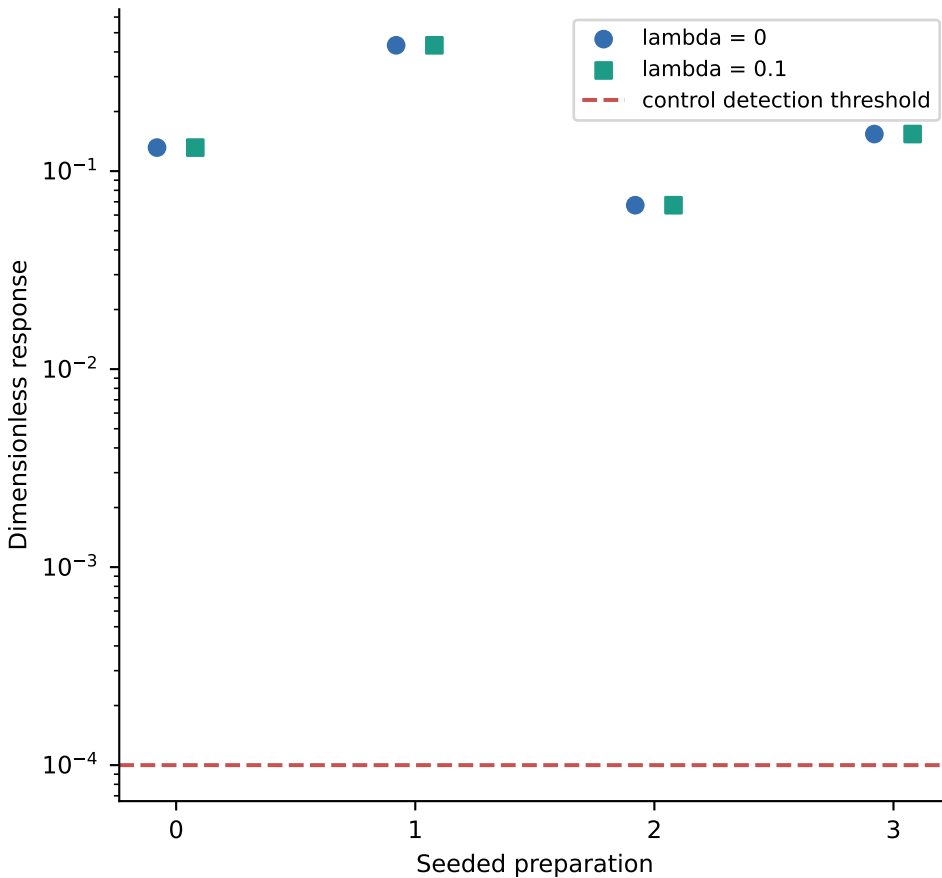
Maximum over every circuit and gate



Independent references and descriptions



Shared-memory order: projector change



Frozen-memory charge defect

